CGIAR
INITIATIVE ON
Diversification in East and Southern Africa

Development of Climate Smart Mechanization and Irrigation Lending Product

January 2024

Contents

Introduction

The Challenge

The importance of agriculture continues to grow across many African countries with agriculture generating up to 50 percent of gross domestic product and contributes over 80 percent to trade in value and more than 50 percent of raw materials to industries (FAO, 2010). Despite the dominance of agriculture on the continent, investment in the sector is still low and limited to export cash crops. Agricultural production has remained static with cereal crop yields and productivity flattening. Agriculture in Africa remains largely dependent on human labor with minimal levels of mechanization. Estimates in Kenya show 2.5 tractors per 1000 hectares compared to South Asia or Latin America with an average of 10 tractors per 1000 hectares. The use of tractors in sub-Saharan Africa remains negligible, shown to barely increase over the past 25 years compared to tenfold increases over the same period in Asia. Food security on the continent is critical with the current agricultural food systems struggling to feed the increased urban population by largely human muscle power.

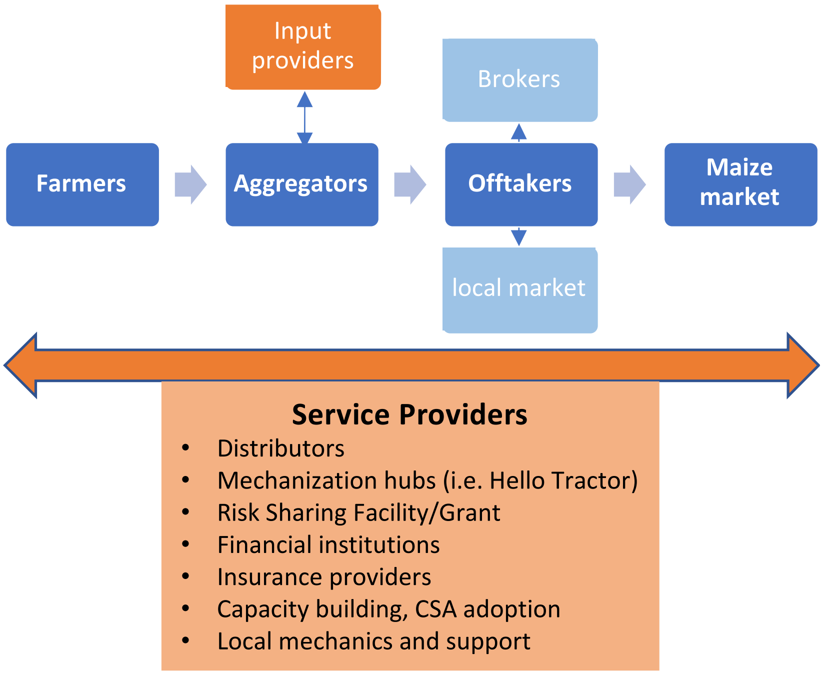
One of the factors contributing to the farmer productivity challenge has been land fragmentation and diminishing farm sizes that inhibits smallholder farmers to be able to economically deploy agri-equipment. African farming systems are almost entirely rainfed and utilize low levels of mechanization. There are numerous proven forms of small-scale irrigation, landscape-level agricultural water management and farming mechanization that are key technologies for smallholders on their pathway out of poverty and food insecurity. However, these technologies though different in form, face similar challenges on scaling and enabling their intended impacts. A key challenge has been the access to structured finance, and the significant capital investment required. Robust market systems are required to provide support for the use and maintenance of the assets, thereby maximizing and extending their utilization.

The high potential of mechanization by smallholders is evidenced by research globally to: 1. Increase farmer profitability, 2. Multiple mechanization solutions and pathways exist to improve nutrition and resilience, 3. Climate smart adaptation and risk mitigation technology can be environmentally sustainable, 4. Scalable smallholder business models exist and 5. Gender equity and inclusivity can be achieved with intentional design. Despite these positive benefits, agricultural mechanization remains inaccessible for most smallholders and adoption barriers persist. Mechanization in agriculture involves more than just giving farmers access to tractors and machinery or providing mechanization services through the public sector. Previous attempts at these approaches have been unsuccessful, with multiple 'pilots' failing to scale to address common regional problems. Scaling is a process of moving towards sustainable systems change at scale, with the framework systematically navigating the complexities involved.

The Proposed Approach

For this concept note, the term 'climate smart assets' refers to mechanisation assets such as irrigation systems, solar water pumps, biodigesters and solar drying kits. The approach adapts the mechanization service provision model used in Zimbabwe and Zambia by Christian Thierfelder and Leon Jamann under the CGIAR Diversification in East and Southern Africa (Work Package One). Their primary goal is to establish a mechanization service provision model and link it to financial partners. Their model goes a step further by supporting capacity building of mechanization service providers such as rural mechanics and manufacturers. The model empowers service providers to make a 30% down payment on the asset/equipment and 70% of the amount to be financed through a loan by a financial institution. The model requires service providers to be bankable and meet the credit scoring and detailed KYC requirements of a financial institution. Without being able to accurately measure the agricultural risk and repayment capacity of the smallholder farmers the loan is priced high.

Structured Value Chain approach



Objective

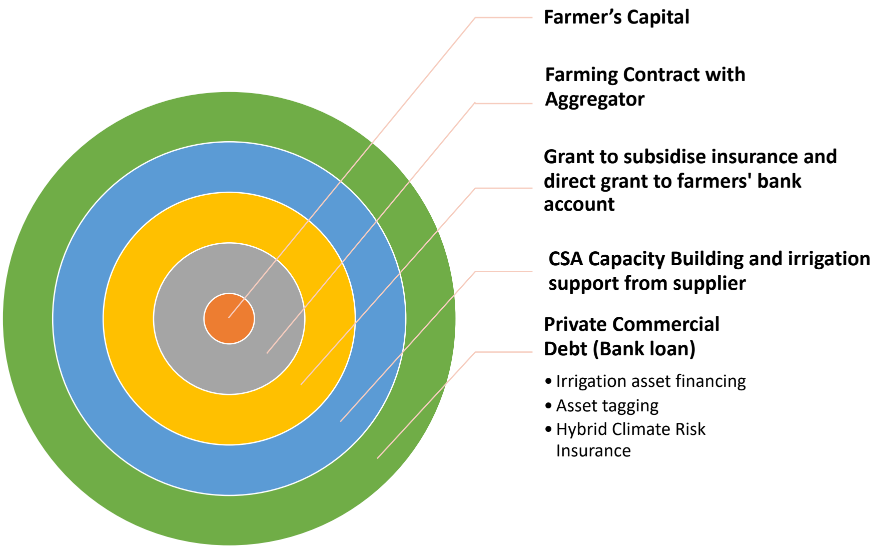
The proposed intervention will operationalise a risk sharing facility that will enable improved access to finance solutions to smallholder farmers for the purchase of climate smart technologies. The facility will cover small ticket size lending for financial institutions, cooperatives and distributors willing to lend to smallholder farmers to help them purchase climate-smart assets. It is crucial to note that the organisation of the farmers is vital to ensure the de-risking and commercial viability of the interventions.

The intervention will cooperate with an array of stakeholders, including financial institutions (FIs) such as Microfinance Institutions and Savings and Credit Cooperatives (SACCOs), farmer cooperatives, Agro-vet retailers, and community-based organizations (CBOs) that have shown interest and meet scaling criteria. In addition, the intervention will engage the services of private sector partners in the distribution and financing of the climate smart assets to ensure sustainable market development.

The concept

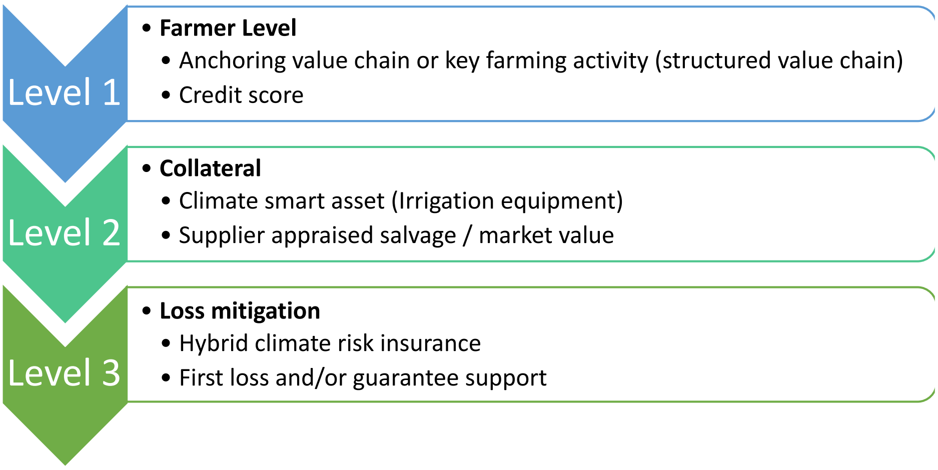
The proposed intervention aims at de-risking access to financing climate smart agricultural assets through extensive technical assistance/capacity building and supporting introduction or development of financial instruments and mechanisms to improve the risk profile for the smallholder farmers for financial institutions. The concept aims to build upon the most effective methods for simplifying financial processes for smallholder farmers in a structured supply chain. This includes exploring options such as farmer cooperatives, where assets are provided to farmers and their instalment payments are collected directly or deducted from the payments made by the cooperative for the delivered produce. Additionally, distributors play a role as sales catalysts, taking on a predetermined percentage of debt responsibility when smallholder farmers face difficulties in covering the costs of the assets.

Bundled Irrigation Asset Loan product



There are three structured levels of comfort from a credit risk perspective, with the first level being the value chain financing viability undertaken by the financial institution. This covers the bundled bespoke climate risk insurance issued by a third-party provider, the contract farming revenue determined by the aggregator and individual credit scoring undertaken by the financial institution. The need for a structured supply chain is crucial in standardising the irrigation credit product and enabling efficient scaling. The second level of comfort is the collateral value of the irrigation asset and true market valuation done by the supplier. The irrigation supplier provides support to the farmer on usage and maintenance to the irrigation asset. The close relationship allows the supplier to be able to accurately calculate the salvage value and effective life of the irrigation asset. Asset tagging and periodical inspections will allow for suppliers to reposes the asset in the case of default. The third level of comfort is the risk sharing facility or grant.

Credit Risk Comfort Levels



The Risk Sharing Facility and/or Farmer Grant

There needs to be an incentive for banks to provide loans to the suppliers of the irrigation assets and support in de-risking their agricultural loan book portfolio. Most smallholder farmers and local irrigation asset suppliers/distributors are deemed high-risk by financial institutions due to the seasonality of their income and lack of pre-requisites (I.e., proof of bankability, institutional framework, collateral). The proposed method is to have an anchoring value chain with a farming contract and support from an aggregator. This ensures that cashflows from the farming activity can be used to assess the repayment capacity of the farmer and asset supplier/distributor creditworthiness.

A risk sharing facility will act as a stimulus to improve lending terms to smallholder farmers as it will cover the risk of credit collection through loss mitigation. A direct grant would be deposited into the accounts of the selected distributors and/or farmers The aim is to support selected local financial institutions to co-create bundled loan products that will be attractive to the smallholder farmers, having suitable loan associated fees, and flexible repayment period to align with seasonal income variations. The grant facility will be intended for smallholder farmers that are involved in commercial farming with contract farming agreements in place. In collaboration with intervention partners, CGIAR will develop a criterion for selecting the farmers and aggregators. The aim is to build upon existing work packages, reduce duplication of data collection and expanding existing business cases to scale.

Capacity Building

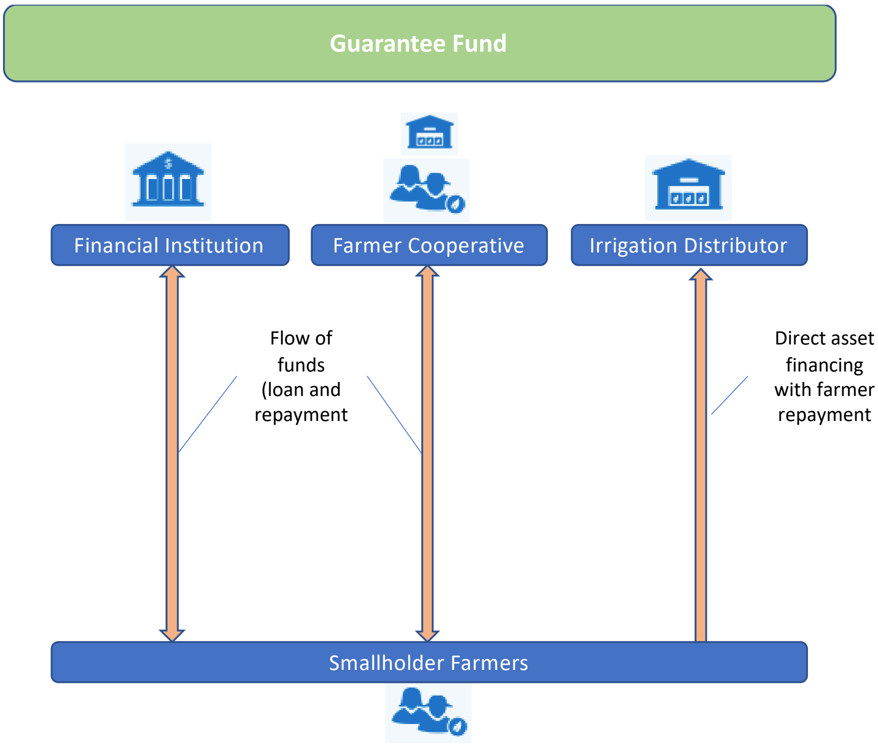
A key barrier in the successful adoption of mechanization in agriculture has been the inadequate training and support to the different actors.

- **Farmers:** The effective use of irrigation assets and adoption of climate smart agricultural practices to mitigate the climate risks being faced will be delivered by agronomists and distributers. The effective use and maintenance of the irrigation assets being deployed is key to ensure adoption and protecting the asset investment. The mechanization adoption will be driven by a needs basis and decisions based on market information. Smallholder farmers will be assisted on being bankable, with financial literacy, climate information systems, market data and basic maintenance of irrigation assets.
- **Financial institutions:** The need to support financial institutions on the effective credit assessment of farmers and suppliers/ distributors. Access to verifiable production data, cost effective know-you-customer (KYC) onboarding, value chain specific cash flow forecasting and assessment of repayment capacity. The aim is to provide Financial Institutions with a cost effective and appropriate means of assessing the credit risk to fairly price the loans. By understanding the internal processes of a financial institution, a portfolio of bankable farmers and distributors can be compiled in accordance with their requirements and supplemented with agronomic data (climate information services, value chain indicators and market scans).
- **Distributors:** To support the development and strengthening of the value chain, the local distributors need to be empowered to fulfil their role. The aim is to create a localised enabling environment to increase the utilization of mechanization that is demand driven and supports the agro-processing industry. The capacity building will aim to strengthen the distributors on their bankability and operations by aiding in their regulatory practices and client support.

Funding Channels

To effectively scale irrigation mechanization requires the use of multiple funding channels and structured options. The three conventional funding channels are the Financial Institutions, Farmer Cooperatives, and Distributors. The initial focus of the intervention will be on the irrigation distribution channel to maximize impact and ensure the demand driven approach is aligned. This funding channel requires the strengthening of the distributor as a key local service provider and empowering them to take co-ownership of the relationship with the farmer.

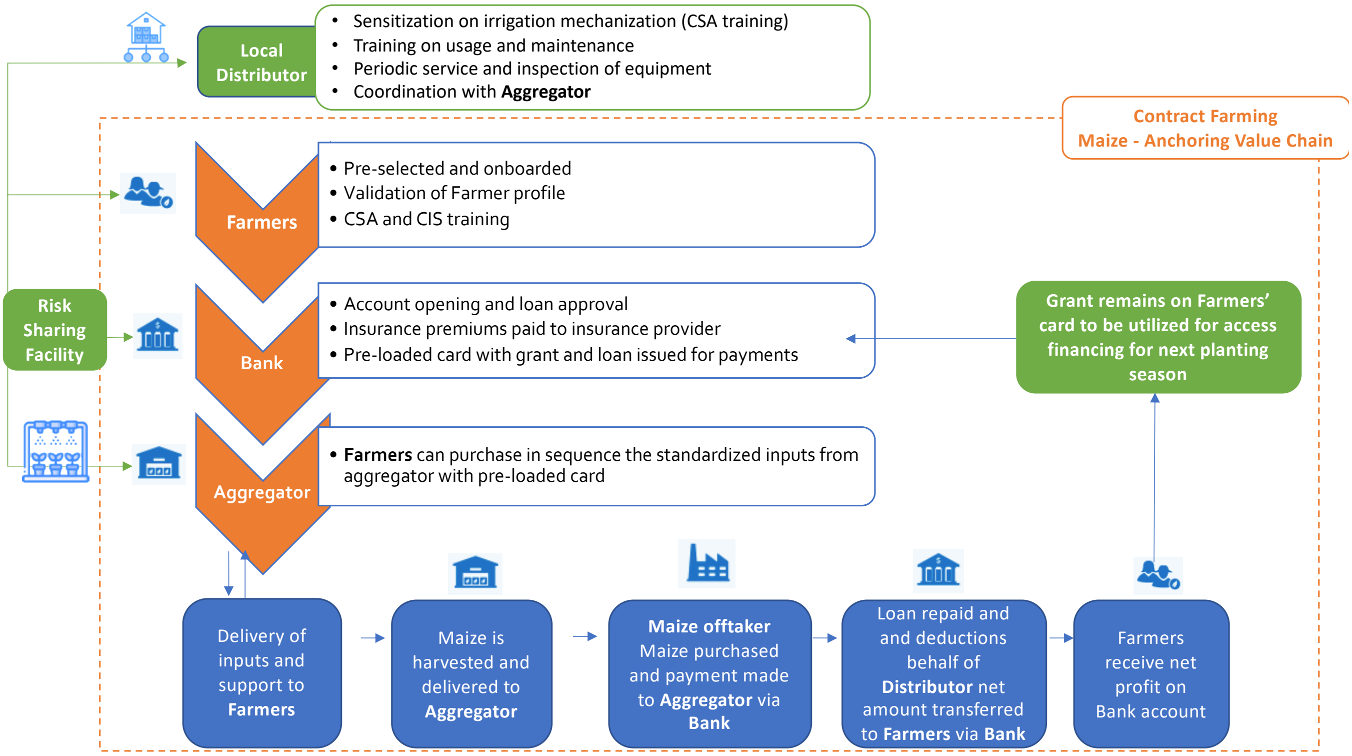
The Three Funding Channels



Channel 1: Irrigation Distributor

By onboarding distributors that have the local network presence, technical capacity and value chain knowledge, the Distributor can be the foundation funding channel. In collaboration with the Aggregator and cooperatives, the Distributor can identify viable farmers and nurture a relationship through capacity building and field demonstrations. Post-sale, the Distributor can account for service and compute the market value of the irrigation asset in the event of a default. Established distributors with a credit system (Bank financing) can act as a sales catalyst with support from the Guarantee Fund/Grant and offer direct asset financing to the farmers. This will increase the adoption of mechanization by effectively catering to the Farmer demands and local market developments.

Overview of irrigation funding integration with Distributor



Channel 2: Financial Institutions

The hight cost and challenges for Financial Institutions to reach smallholder Farmers and accurately assess their credit risk remains a challenge in most rural communities. Many Farmers are deemed un-bankable and revert to informal financial banking services to meet their needs. This funding channel will look at the bankability and business case development of service providers and Distributors in the value chain. The Risk Sharing Facility/Grant will ensure that credit flows from the Financial Institution via the Distributor to the Farmers. By setting a percentage for the Facility/Grant that covers 25-65% of the loan amount in the case of default, the Financial Institution can further de-risk and closely monitor the market value of the asset deployed.

Channel 3: Farmer Cooperatives and Aggregators

Cooperatives have established systems in place to disburse payments to their members and aggregate farmers for CSA capacity building within the community. The systems allow for saving, loan repayment and working capital advances. The Risk Sharing Facility/Grant can be provided to the Cooperatives, that would in turn extend asset financing for the irrigation equipment. An alternative model would be where the Aggregators are the intermediaries between the Farmers and Financial Institutions and Distributors. With a farming contract and anchoring value chain, the Farmer would receive deductions from his existing payments from their yield payments per harvest. The intervention will aim to identify which options are most viable within the context and provide active support towards its development.

Actors to be Engaged

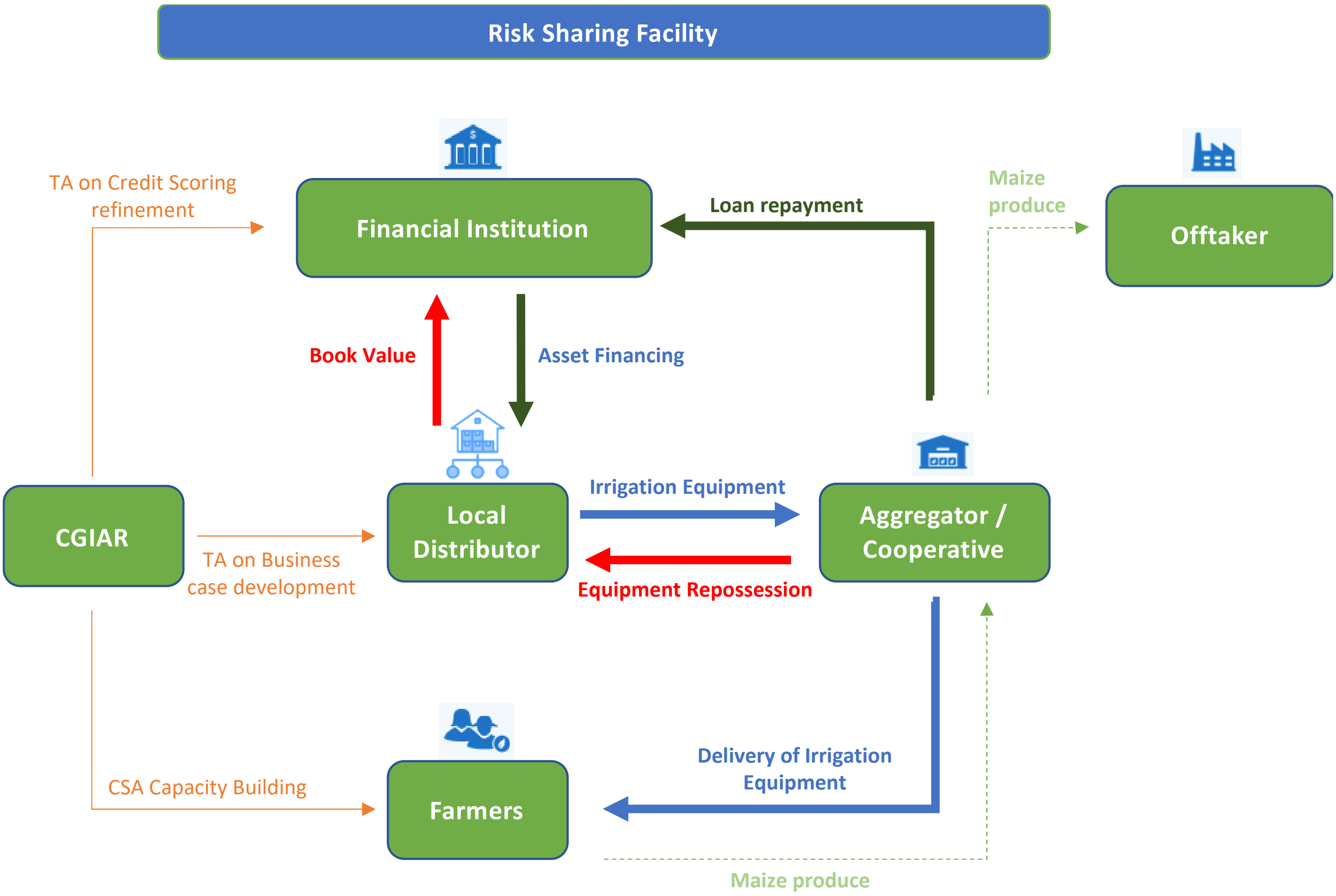
Actor	Role	Revenue model	Incentive to Participate
Irrigation Supplier	• Sell irrigation equipment/ assets to Distributors • Provide training on equipment use and maintenance to Distributors	Margin on irrigation equipment sales	• Grow customer base and market • Increase sales of irrigation equipment
Irrigation Distributor	• Sell irrigation equipment to Farmers • Educate community of irrigation and mechanization benefits • Provide training on equipment use and maintenance to Farmers • Provide asset financing to Farmers for irrigation equipment • Service and value equipment in the case of default and repossession	Margin and commission on irrigation equipment sales	• Customer retention • Increase sales and presence • Expand distribution network and product range
Financial Institution	• Providing loans to Distributors, Farmers and other value chain actors (Agri-entrepreneurs) • Increase customer base • Upsell financial products (household and education loans, savings products)	Interest on loans	• Increased loan disbursements and revenues • Improved access to data to attract new agri customers
Insurance provider	• Co-design and deploy hybrid climate risk insurance product to Farmers	Insurance premiums	• Increased sales of insurance products • Access to production and farmer data to refine products
Aggregator	• Farmer aggregation, coordination and value chain rigidity • Contract farming lead • Capacity building (CSA adoption) lead • Key in creating demand of Irrigation equipment and mechanization • Scaling partner	Margin on crop sales	• Increase quality and yield of crop supply • Contributing to alleviation of poverty in rural communities

Implementation Approach

Inception Phase

An inception meeting will be held to have a detailed review of the scope and workplan with various actors in the financial sector, cooperatives, and private sector to inform design of risk sharing facility structure, design of technical assistance requirements for various stakeholders, identifying suitable partners in providing this TA, identifying, and contracting of actors across funding channels and TA activities.

Proposed structure



The goal will be to design and co-create an integrated irrigation financing product that scales across the three financing channels. The product would be demand-driven by building on an existing contract farming scheme in a structured value chain with aggregators, that advise on the investment case for irrigation equipment and the repayment capacity of the individual Farmers.

Needs Assessments and Scheme Development:

This phase will involve identifying and analysing prospective actors from the different channels to gain an understanding of their operations and experiences, financing needs and capacities, willingness to collaborate and key conditions to participate in the financing scheme. A review of prequalified and selected service providers from the pilot phase of the mechanization financing work package.

- **Analyze the capacity** of the target Farmers and the business case viability of the Distributors in the maize value chain within the priority counties vis-à-vis the costs of the appropriate irrigation assets the potential to take up external financing and repay it. This would also help ascertain the viable value chains and smallholder farmer typology that can feasibly participate in the intervention.

- **A review of the Zambian Market and Supply chain analysis of maize and supplementary value chain.**

1. Full review of current existing sector and supply chain studies relevant to the geographic area targeted at county level, as well as any relevant data available.
2. Review published articles and findings by past studied undertaken by CIMMYT and CGIAR.
3. Comparative of 475 prospective service providers for mechanization identified (review scoring data points).

- **Supply Chain Partner Engagement**

1. Shortlist of supply chain actors that will be approached for commitment to and participation in this initiative
2. Assessment and validation of supply chain partners on of their viability for the pilot
3. Mapping of impact of market risk and climate change-related factors (drought, flooding and other) on their smallholder supply base will be assessed and quantified.

- **Business and Investment case review**

A refinement of the Business Cases by concluding on the Supply chain mapping (ii) intercropping analysis (iii) location and competitive environment (iv) role of insurance and certification (v) existing available credit support, incl. government loan, guarantee and other schemes (vi) credit, market and E&S risk analysis (vii) offtake commitment, pricing and other (contractual) arrangements between farmers, traders, mills and other supply chain actors (viii) long-term and short-term financing needs.

- **Design and implementation**

1. Supply chain stakeholder roundtable to align on scope and roles.
2. Selection of location with support of value chain aggregator and standardize support structure for scalability.
3. Coordinate Aggregator engagement with selected service providers and input suppliers.
4. Final dedicated pre-paid card developed, and point-of-sale device delivered to service providers.
5. MoU's and partnership agreements finalised
6. Contracts with offtakers and aggregators confirmed with set pricing.
7. Prices with service providers fixed for planting cycle
8. Grant provision confirmed or Risk Sharing Facility executed.
9. Indicative loan and insurance term sheet

If a contract farming scheme is not in place with a structured flow of inputs and support from the aggregator the following actions will be undertaken:

- **Farmer profiling**

The intervention will leverage CGIAR's climate credit risk scoring model (SF One) to assess the repayment capacity of farmers within the cooperatives against key credit risk drivers as well as the Know-Your-Client requirements of the partner financial institutions. Using in-depth farmer information, CGIAR will analyze the cash flow and repayment capacity of small-scale producers and generate individual credit scores for each farmer. These credit scores help to identify key credit risks and assist financial institutions to develop lower risk agricultural lending portfolios. The smallholders, who are determined to be 'acceptable' by the financial institutions based on their internal risk/return parameters and the credit risk scores, will be grouped into a portfolio of bankable farmers ready to receive their assets on credit. For the farmers who are deemed not to qualify for a loan at once, a separate analysis will be conducted to identify the main hurdles to their bankability status.

- **Scaling pilot and Mobilization of Farmers**

1. Harmonized data collection (KYC, farmer profile, production data, capacity building scope etc)
2. Baseline study ahead of targeted planting cycle
3. Sensitization of farmers on the scheme and irrigation equipment
4. CSA capacity building provided and TA to Distributors
5. Onboarding of farmers and linkages to Distributors
6. Loan disbursed and accessible by Farmers through the pre-loaded card in line with planting cycle
7. Monitoring of farming progress done bi-monthly by aggregator agents and agronomists.
8. Training coordination for the different planting phases.

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